DATABASE MANAGEMENT SYSTEMS / CASE STUDY • REVIEW 1

FraudLens

Financial Transaction & Fraud Investigation Management System

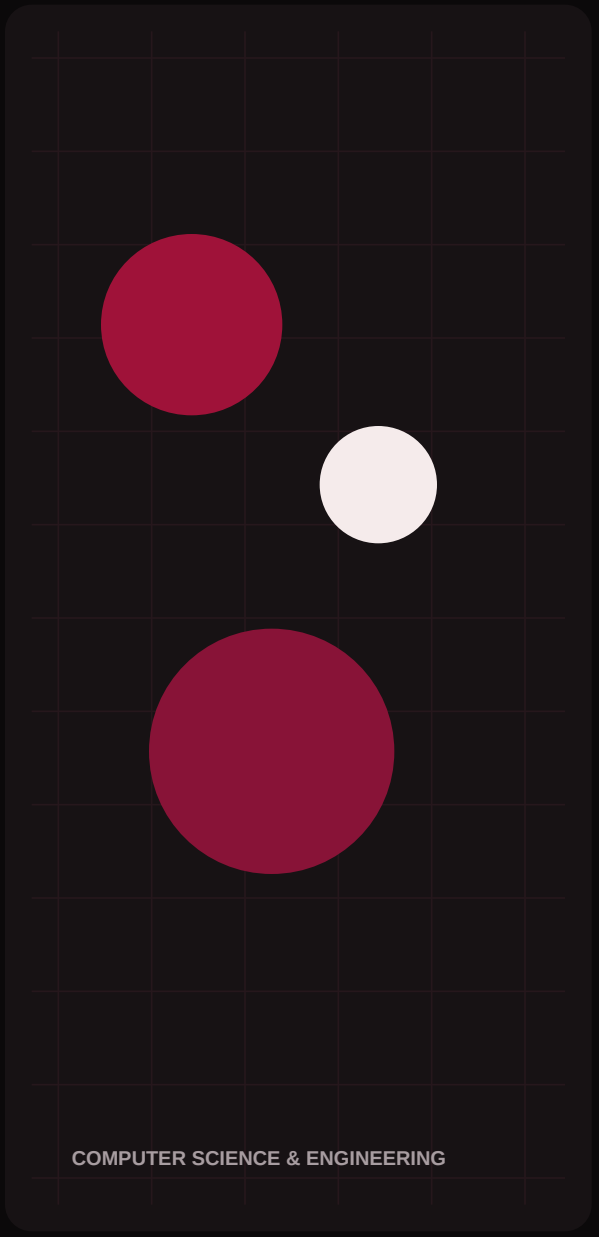
Relational domain modelling, 3NF architecture, Oracle SQL DDL with named constraints and sequences, controlled DML, and analytical queries.

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Review 1 / Relational Schema Design & SQL Foundations

BCSE304L Database Management Systems • SCOPE, VIT Vellore



Fraud data scattered across silos

Banks move tens of thousands of transactions a minute: RTGS, NEFT, IMPS, ATM, POS, UPI.

THE CONTEXT

Fragmented storage

Customers, ledgers, transactions and case notes sit in separate spreadsheets with no foreign-key links.

THE RISK

Drift + weak validation

Duplicated demographics cause update anomalies; negative balances and orphan tickets slip through.

OUR RESPONSE

One 3NF Oracle schema

Six core entities with PK/FK, cascading deletes and engine-level CHECK constraints enforce valid ranges and states.

RESEARCH QUESTION

How can a bank unify customers, accounts, transactions, alerts and investigations so suspicious activity is traceable and data stays correct?

Scope: Review 1 covers relational requirements, ER model, 3NF mapping, Oracle DDL, controlled DML, and baseline analytical queries. Review 2 integrates procedural surveillance, views, stored procedures, functions, triggers, and full audit logs.

Three failures, three fixes

Operational deficiencies mapped to the FraudLens relational solution.

Operational deficiency	Impact on banking operations	FraudLens relational solution
Fragmented storage	Transactions, alerts and case notes live in disparate systems without foreign-key links.	One centralised Oracle schema with strict primary/foreign key relationships.
Redundancy & drift	Customer demographics duplicated across spreadsheets cause update anomalies.	Normalised to 3NF; each customer attribute is stored exactly once.
Weak validation	Spreadsheets allow negative balances, zero amounts and orphan investigation tickets.	Engine-level CHECK constraints enforce valid ranges, balances and statuses.

Engine Enforcement Principle: Every validation rule is enforced inside the Oracle database engine via constraints and triggers, not left to application code.

What the database must enforce

Functional objectives on the left; enforced rules BR-01 to BR-06 on the right.

FUNCTIONAL OBJECTIVES

- **Account hierarchy:** One customer may hold multiple accounts across Savings, Current, Salary, and NRI types.
- **Transaction journal:** Captures type (DEPOSIT, WITHDRAWAL, TRANSFER), timestamp, channel, counterparty, and status.
- **Rule-based alerts:** Dynamic severity scoring across LOW, MEDIUM, HIGH, and CRITICAL brackets.
- **Case lifecycle:** Structured progression through OPEN, UNDER REVIEW, and CLOSED with assigned investigators.
- **Analytical querying:** Comprehensive baseline reporting using character formatting, numeric filters, group metrics, and joins.

ENFORCED BUSINESS RULES

BR-01	Each account belongs to exactly one customer (ON DELETE CASCADE).
BR-02	Account balance must stay strictly non-negative (CHECK balance >= 0).
BR-03	Transaction amount must be strictly greater than zero (CHECK amount > 0).
BR-04	A fraud alert references an existing transaction; cascade deletes on purge.
BR-05	An investigation links one alert to one active investigator.
BR-06	Transactions above INR 50,000 are flagged for compliance review.

ER model of the implemented schema

Six entities designed in Oracle SQL Developer Data Modeler; P = primary, F = foreign, U = unique key.

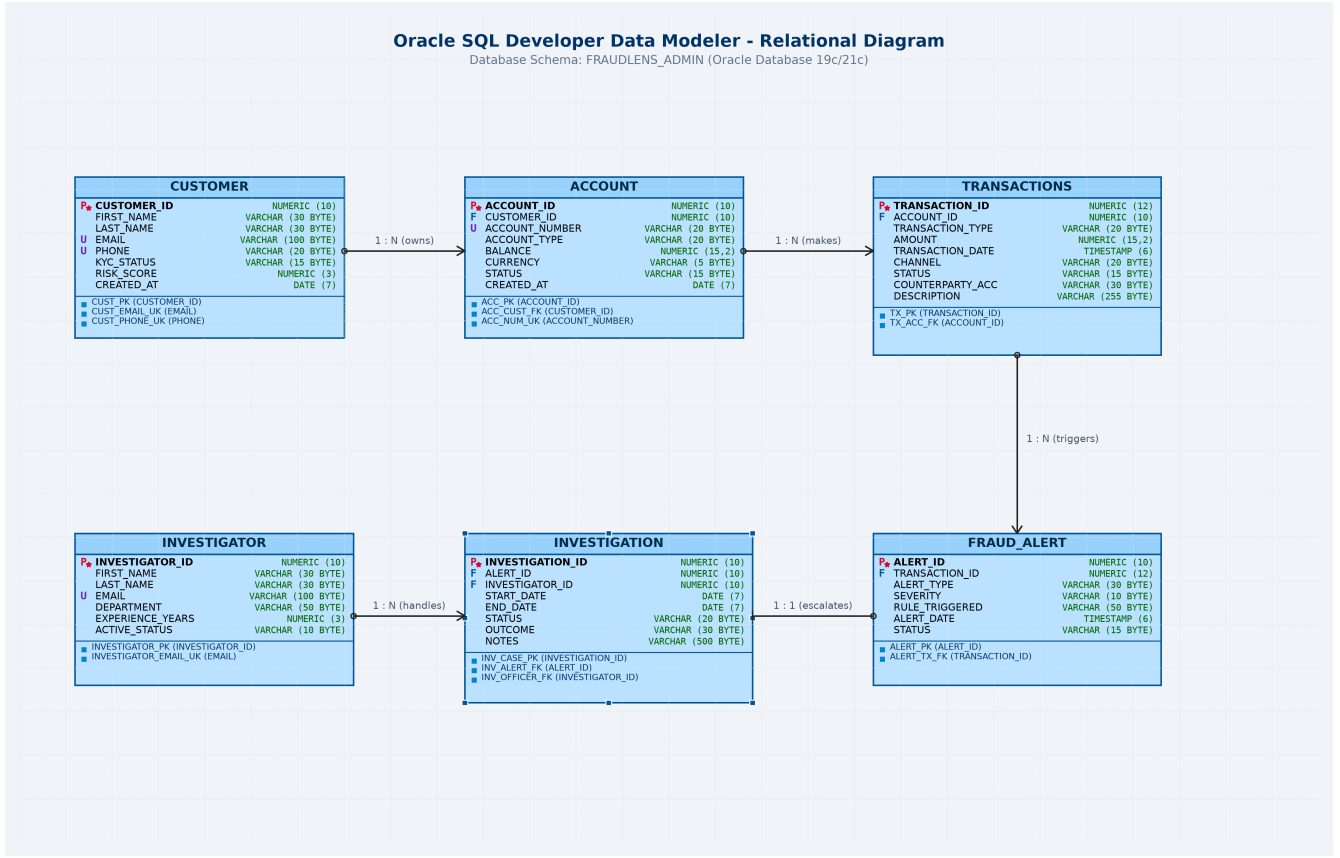


Figure 1: FraudLens relational diagram (Oracle SQL Developer Data Modeler / Oracle Database 23ai)

Keys and constraints encode the rules

Single-column surrogate keys; every table verified against 1NF, 2NF and 3NF.

Table	Primary key	Foreign key(s)	Key constraints enforced
CUSTOMER	`customer_id`	-	email, phone UNIQUE NOT NULL; kyc_status CHECK; risk_score 0-100
ACCOUNT	`account_id`	`customer_id` → `CUSTOMER`	account_number UNIQUE; balance >= 0; type and status CHECK
TRANSACTIONS	`transaction_id`	`account_id` → `ACCOUNT`	amount > 0; transaction_type, channel, status CHECK
FRAUD_ALERT	`alert_id`	`transaction_id` → `TRANSACTIONS`	severity CHECK (LOW to CRITICAL); status DEFAULT 'PENDING'
INVESTIGATOR	`investigator_id`	-	email UNIQUE NOT NULL; experience_years >= 0; active_status CHECK
INVESTIGATION	`case_id`	`alert_id, investigator_id`	status CHECK; outcome CHECK; end_date >= start_date

1NF

Atomic scalar values

All table attributes store atomic scalar values. No repeating groups or multi-valued phone lists; first and last names separated.

2NF

No partial dependencies

Every relation relies on single-column surrogate primary keys. No non-prime attribute depends on a proper subset of any candidate key.

3NF

No transitive dependencies

Non-key columns depend exclusively on primary keys. Investigator department and customer KYC status live only in home entities.

Oracle DDL with named constraints

From sql/01_schema.sql; named constraints give deterministic error diagnostics.

CREATE: ACCOUNT (DDL Snippet)

```
CREATE TABLE ACCOUNT (
  account_id   NUMBER(10) CONSTRAINT pk_account PRIMARY KEY,
  customer_id  NUMBER(10) NOT NULL
               CONSTRAINT fk_acc_customer
               REFERENCES CUSTOMER(customer_id)
               ON DELETE CASCADE,
  account_number VARCHAR2(20)
               CONSTRAINT uq_acc_number UNIQUE NOT NULL,
  account_type VARCHAR2(20) DEFAULT 'SAVINGS'
               CONSTRAINT chk_acc_type CHECK (account_type
               IN ('SAVINGS','CURRENT','SALARY','NRI')),
  balance      NUMBER(15,2) DEFAULT 0.00
               CONSTRAINT chk_acc_balance CHECK (balance >= 0),
  status       VARCHAR2(15) DEFAULT 'ACTIVE'
               CONSTRAINT chk_acc_status CHECK (status IN
               ('ACTIVE','FROZEN','DORMANT','CLOSED'))
);
```

CREATE: TRANSACTIONS (Abbreviated)

```
CREATE TABLE TRANSACTIONS (
  transaction_id NUMBER(12) CONSTRAINT pk_transactions PRIMARY KEY,
  account_id     NUMBER(10) NOT NULL
                 CONSTRAINT fk_tx_account
                 REFERENCES ACCOUNT(account_id) ON DELETE CASCADE,
  amount         NUMBER(15,2) NOT NULL
                 CONSTRAINT chk_tx_amount CHECK (amount > 0),
  channel        VARCHAR2(20) NOT NULL
                 CONSTRAINT chk_tx_channel CHECK (channel IN
                 ('ATM','POS','NET_BANKING','MOBILE_APP','UPI'))
);
```

SEQUENCES

Six PK sequence generators

START WITH sits above sample IDs (e.g., customer sequence starts at 1011 after sample 1001-1010; transaction sequence at 50019 after sample 50001-50018) to avoid key collision during execution.

Safe schema changes, reversible DML

From sql/02_ddl_dml_demos.sql; structural changes on a staging table, updates under a savepoint.

ALTER / DROP: STAGING TABLE

```
-- 1. Schema Modification on Staging Table
ALTER TABLE FRAUD_RULE_CONFIG
  ADD (rule_description VARCHAR2(250));

ALTER TABLE FRAUD_RULE_CONFIG
  MODIFY (rule_code VARCHAR2(60));

ALTER TABLE FRAUD_RULE_CONFIG
  DROP COLUMN rule_description;

DROP TABLE FRAUD_RULE_CONFIG PURGE;
```

DML UNDER SAVEPOINT

```
-- 2. Transaction Control with Savepoint
SAVEPOINT dml_demo_start;

UPDATE CUSTOMER
  SET kyc_status = 'VERIFIED', risk_score = 5
  WHERE customer_id = 9999;

DELETE FROM CUSTOMER
  WHERE customer_id IN (9998, 9999);

ROLLBACK TO dml_demo_start;
```

WHY A SAVEPOINT?

Baseline stays pristine

The rollback restores the 56-record baseline dataset immediately after the demonstration, ensuring that subsequent analytical queries, joins, and test cases run against known, predictable data.

Single-row functions

UPPER, LOWER, LENGTH and SUBSTR standardise customer profiles.

SQL*Plus - FRAUDLENS_USER@localhost:1521/freepdb1

```
SQL> SELECT customer_id,
  2         UPPER(first_name || ' ' || last_name) AS full_name_uppercase,
  3         LOWER(email) AS standardized_email,
  4         LENGTH(phone) AS phone_len,
  5         SUBSTR(phone, 1, 3) AS country_code
  6  FROM CUSTOMER
  7  WHERE customer_id IN (1001, 1002, 1005);
CUSTOMER_ID FULL_NAME_UPPERCASE    STANDARDIZED_EMAIL          PHONE_LEN COUNTRY_CODE
----------- ---------------------- --------------------------- ---------- ------------
       1001 AARAV SHARMA           aarav.sharma@finmail.in            13 +91
       1002 PRIYA NAIR             priya.nair@corpmail.com            13 +91
       1005 VIKRAM SINGHANIA       vikram.s@apexholdings.in           13 +91

3 rows selected.
```

*Figure 2: Single-row character functions with query results (Executed on Oracle 23ai Free / FREEPDB1 via SQL*Plus)*

IN, BETWEEN and ORDER BY

High-value digital-channel transactions between INR 50,000 and 300,000.

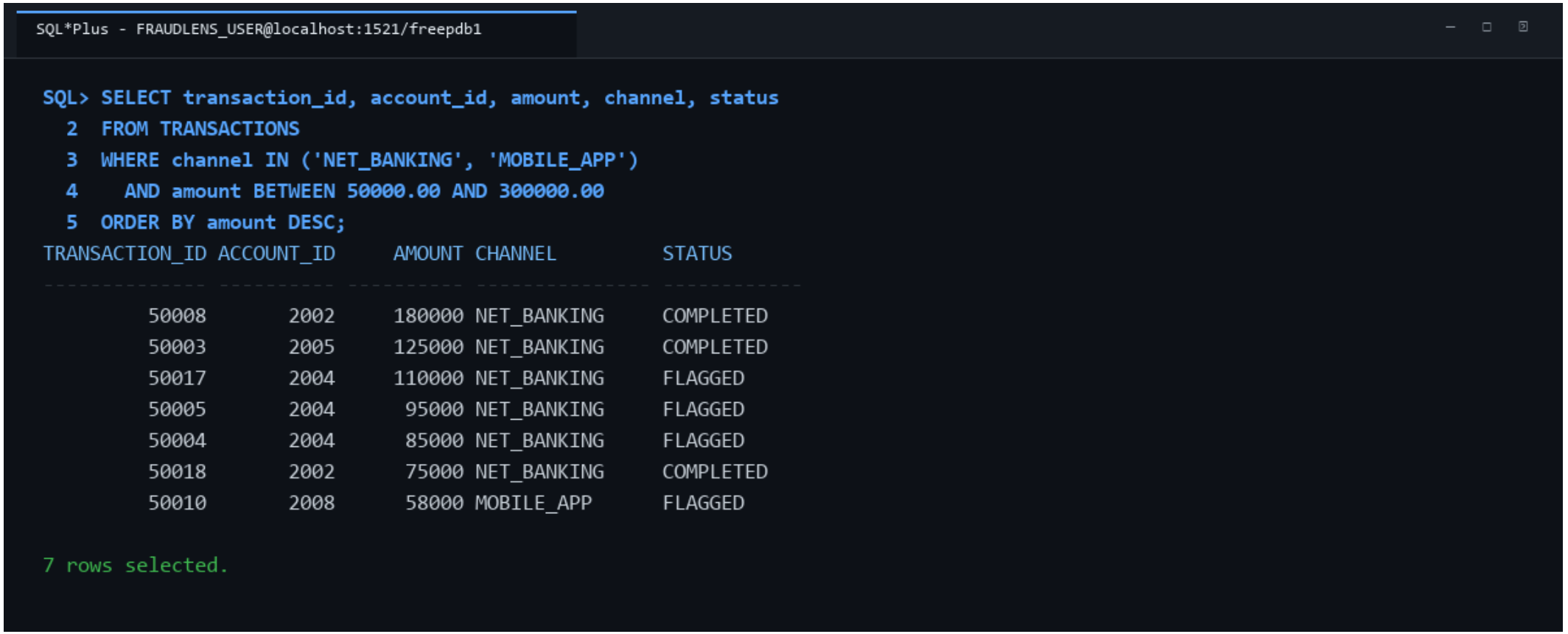


*Figure 3: Range filtering with IN, BETWEEN and descending sort (Executed on Oracle 23ai Free / FREEPDB1 via SQL*Plus)*

GROUP BY and HAVING

Transaction count, turnover and average ticket per channel, channels with 3+ transactions.

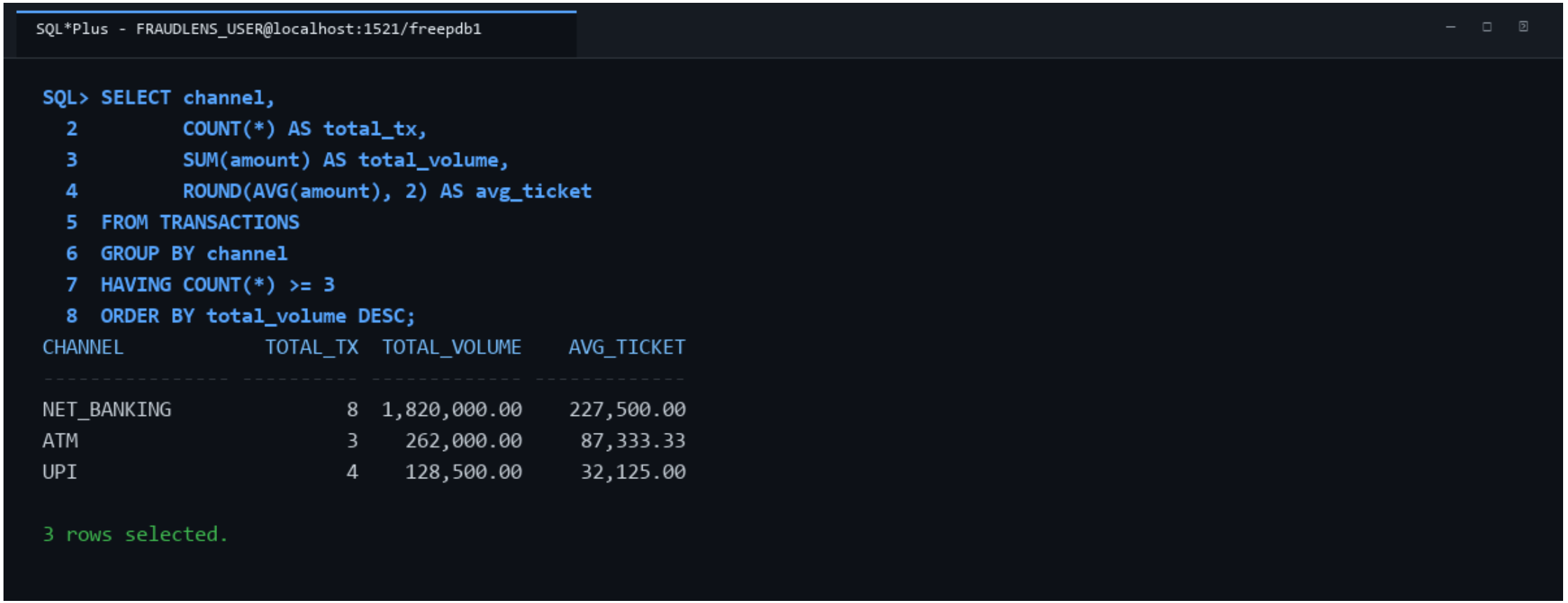


*Figure 4: Channel aggregation metrics (Executed on Oracle 23ai Free / FREEPDB1 via SQL*Plus)*

Four-table analytical join

Customer, account, transaction and alert linked to surface high-risk exposure.

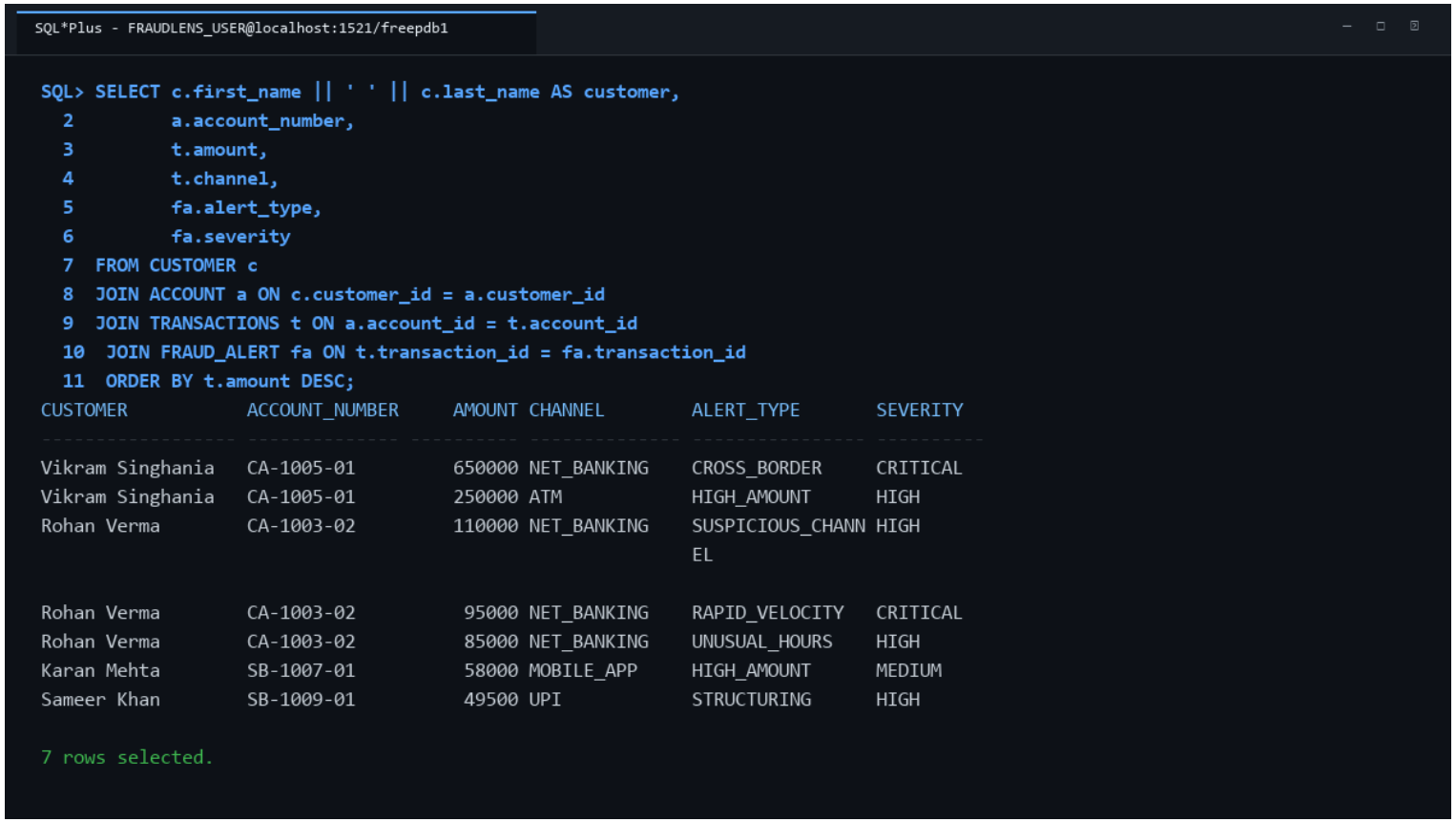
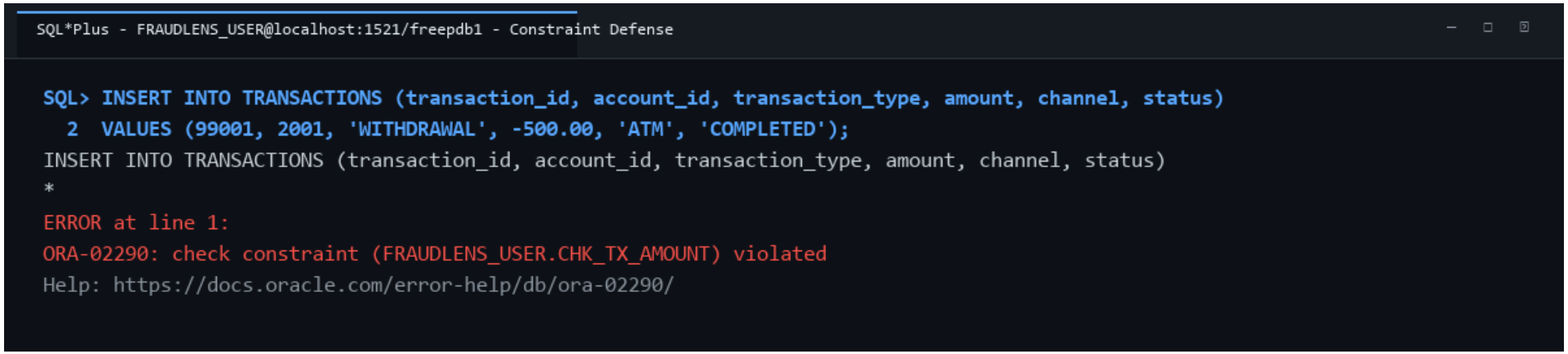


*Figure 5: Four-table join tracking high-risk alerts to accounts (Executed on Oracle 23ai Free / FREEPDB1 via SQL*Plus)*

Constraints reject bad data

Negative tests trigger engine-level exceptions; positive tests are accepted.



Type	Constraint	Input tested	Expected	Status
Negative	`CHECK (amount > 0)`	amount = -500.00	Rejected, ORA-02290	PASSED
Negative	`UNIQUE (email)`	existing aarav.sharma@finmail.in	Rejected, ORA-00001	PASSED
Negative	`FOREIGN KEY (customer_id)`	customer_id = 999999	Rejected, ORA-02291	PASSED
Positive	`CHECK (kyc_status)`	kyc_status = 'VERIFIED'	Accepted and updated	PASSED
Positive	`CHECK (risk_score)`	risk_score = 85	Accepted and stored	PASSED

Foundation first, procedures next

Phase 1 delivers a verified 3NF base; Phase 2 adds procedural surveillance.

01 INTEGRITY

Zero orphan records

Cascading FKs ensure no orphan transactions, alerts or cases can exist.

02 NO ANOMALIES

Strict 3NF structure

Customer and investigator details live in one authoritative relation.

03 DOMAIN DEFENCE

Database-level CHECKs

Named CHECKs block negative balances, zero amounts, and invalid statuses.

Review 2 component	Planned implementation	Objective in FraudLens
Advanced subqueries	Correlated and multi-row (IN, ANY, ALL)	Flag accounts far above their own historical average
Database views	VW_HIGH_RISK_TRANSACTIONS, VW_INVESTIGATION_SUMMARY	Dashboards without complex application joins
PL/SQL procedures	SP_ASSIGN_INVESTIGATOR and case progression	Validate constraints before state transitions
User-defined functions	FN_GET_TOTAL_TX_AMOUNT, FN_CALCULATE_RISK_TIER	Dynamic customer risk tiers and account volume scoring
Triggers & audit logs	AFTER UPDATE row-level triggers to audit tables	Non-repudiation and forensic case lifecycle logging

FRAUDLENS • REVIEW 1 COMPLETE

Model correctly. Enforce in the engine.

Keys and constraints protect relationships; queries turn clean data into fraud signals.

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Thank you / Questions